PILOT: Inter-individual Differences in Susceptibility to Cyber-manipulation: A Multi-agent Simulation Approach with High-dimension State Space of Political Opinions

Stijn Van Severen[1,*] Thomas De Schryver[1]

[1] Ghent University

[*] Corresponding author

March 23, 2026

Abstract

This pilot examines how inter-individual differences moderate the effectivity of cyber-manipulation on political opinions using an ontology-driven multi-agent simulation pipeline. Run 6 uses an attacked-only profile-panel design in which one fixed misinformation leaf is applied to each pseudoprofile across multiple political-opinion leaves. Attack effectivity is defined as within-profile post-minus-baseline movement, with absolute shift $A_{ik} = |\text{post}_{ik} - \text{baseline}_{ik}|$ as the primary outcome so that issue-specific signed movements do not cancel out. The pilot generated 100 pseudoprofiles, 400 attacked opinion rows, and 4 repeated attacked outcomes per profile. Mean absolute attacked shift was 43.35, mean signed shift was 22.86, mean attack realism was 0.71, and mean post-exposure plausibility was 0.73. The structural model is now a repeated-outcome path SEM in which multiple attacked opinion deltas are regressed on profile moderators simultaneously. To compute a profile-level susceptibility summary without inheriting multicollinearity from a small-sample OLS, a cross-validated ridge aggregation is fit after the run and converted into a post hoc empirical susceptibility index. The pilot should therefore be read as methodological validation and exploratory evidence rather than as claim-ready causal estimation.

1 Introduction

Political influence campaigns increasingly operate in the space between mass communication, personalized persuasion, and platform-native misinformation. Rather than relying on technical compromise alone, they often work through selective framing, pseudo-consensus, strategic ambiguity, distrust activation, and identity-congruent narrative design (Bennett & Livingston, 2018; Lazer et al., 2018; Lewandowsky et al., 2017; Pennycook & Rand, 2019, 2021; Vosoughi et al., 2018). This makes the key research problem not only whether content is false, but whether the cognitive conditions of judgement are being shaped in ways that weaken self-directed political belief formation.

That concern is closely related to recent discussions of cognitive warfare and cognitive superiority in security policy. NATO-oriented analyses increasingly frame the contested terrain as one in which human attention, interpretation, trust, and decision-making become operationally relevant dimensions of conflict and competition (Johns Hopkins University & Imperial College London, 2021; Hung & Hung, 2022; Paulauskas, 2024). The present study adopts *cognitive sovereignty* as the substantive framing for whether political opinion formation remains self-directed under digital adversarial pressure.

The research question is straightforward: *How do inter-individual differences moderate the effectivity of cyber-manipulation on political opinions?* The methodological challenge is less

straightforward. A real-world dataset that cleanly aligns profile structure, attack-vector semantics, repeated political-opinion items, and post-exposure response states is difficult to obtain, ethically constrained, and often poorly matched to ontology-level modeling. This makes a transparent, auditable simulation architecture useful as a methodological instrument, provided the architecture is explicit enough to be criticized and improved (Cioffi-Revilla, 2002; Rosseel, 2012).

2 Materials and Methods

2.1 Ontology-Constrained Pilot Design

Run 6 uses three hierarchical ontologies: PROFILE, ATTACK, and OPINION. The ontologies remain hierarchical in storage, but estimation is leaf-based. PROFILE combines continuous leaves (e.g., age and Big Five means) with categorical leaves (e.g., sex). ATTACK contributes one fixed pilot leaf, `ATTACK_VECTORS > Social_Media_Misinformation > Misleading_Narrative_Framing`. OPINION contributes repeated leaf nodes from the focused domain Defense_and_National_Security.

The design is attacked-only. Each pseudoprofile is crossed with multiple opinion leaves, so one fixed attack leaf is connected to multiple attacked deltas for the same individual profile. This architecture is more coherent with the moderation question than a treatment-versus-control setup when the research focus is heterogeneity of attack effectivity under exposure rather than incremental exposure effects relative to no exposure.

2.2 Baseline, Exposure, and Post-Exposure Assessment

For each profile i and opinion leaf k, the pipeline elicits a baseline opinion and a post-exposure opinion on a signed integer scale from -1000 to $+1000$. Exposure texts are generated as realistic social-media-native manipulative messages, then audited for realism. Post-exposure responses are audited for coherence and boundedness. This means every analyzed row is not merely a model output, but a generated output that has passed a second review layer designed to detect implausible reversals, coarse rounding, and issue-incoherent response shifts.

2.3 Effectivity Outcome and Profile-Level Susceptibility

The signed attacked shift for profile i on opinion leaf k is

$$\Delta_{ik} = \text{post}_{ik} - \text{baseline}_{ik}, \tag{1}$$

and the primary effectivity outcome is the absolute attacked shift

$$A_{ik} = |\Delta_{ik}|. \tag{2}$$

The profile-level observed effectivity summary is the mean attacked shift magnitude

$$E_i = \frac{1}{K} \sum_{k=1}^{K} A_{ik}, \tag{3}$$

where K is the number of repeated opinion leaves for profile i.

The repeated-outcome SEM is a profile-level path model:

$$A_{ik} = \alpha_k + X_i\beta_k + C_i\gamma_k + \varepsilon_{ik}, \tag{4}$$

where each repeated attacked opinion outcome A_{ik} is regressed on the same profile vector X_i and control vector C_i, while residual covariance between attacked opinion leaves is estimated within the SEM.

The post hoc empirical susceptibility index is not an input variable. It is derived after estimation from a target-conditional regularized aggregation. For each configured attack-opinion task $t \in \mathcal{T}$, a ridge model is fit:

$$\hat{A}_{it} = \hat{\beta}_{0t} + \sum_j \hat{\beta}_{jt}^{(\text{ridge})} X_{ij}, \tag{5}$$

and the conditional susceptibility score is then aggregated as

$$S_i^*(\mathcal{T}) = \text{PctRank}\left(\sum_{t\in\mathcal{T}} w_t \hat{A}_{it}\right), \qquad w_t \propto \frac{n_t}{\text{CV–MSE}_t}. \tag{6}$$

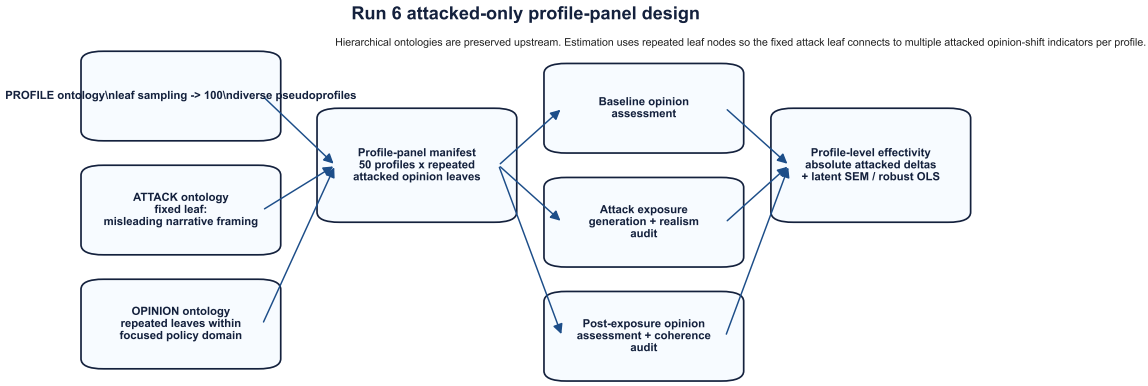
This index is therefore descriptive. It summarizes the model-implied profile propensity toward larger attack effectivity for the configured attack/opinion target set while shrinking unstable coefficient combinations.

3 Results

Run 6 generated 100 pseudoprofiles and 400 attacked opinion rows. The fixed attack leaf was linked to 4 repeated opinion indicators per profile. Mean absolute attacked shift was 43.35 and mean signed shift was 22.86. Mean attack realism was 0.71, while mean post-exposure plausibility was 0.73.

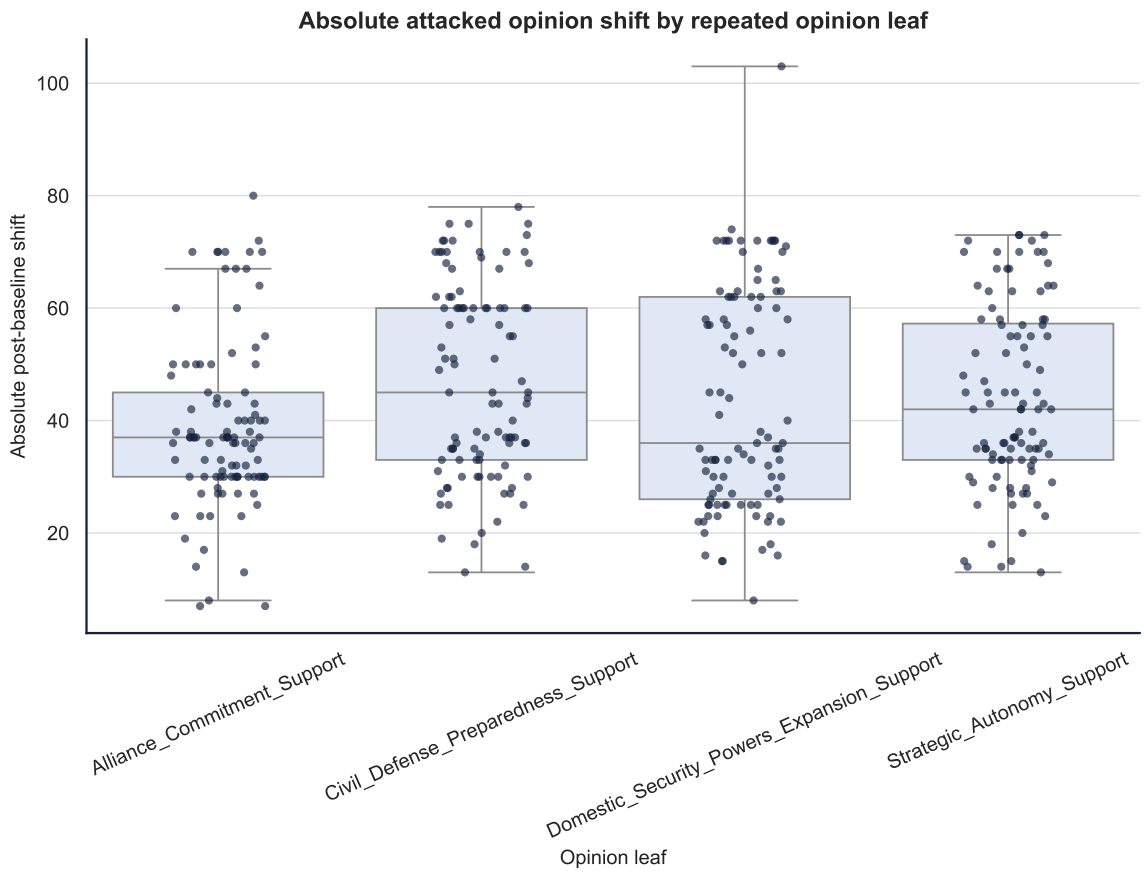
The measurement-and-moderation structure is summarized in Figure 1. Figure 2 shows the distribution of absolute attacked deltas across repeated opinion leaves. Figure 3 shows the descriptive susceptibility weights that summarize the post hoc profile index, and Figure 4 shows the repeated-outcome path-SEM coefficient matrix. The strongest normalized moderator weights in run 6 were Sex Male (8.5% of normalized moderator weight); Age Years (8.5% of normalized moderator weight); Sex Female (6.9% of normalized moderator weight); Big Five Neuroticism Impulsiveness % (6.0% of normalized moderator weight). At the ontology-group level, the largest aggregated weight shares were Personality: Extraversion (20.6%); Demographics: Sex (18.2%); Personality: Openness (15.7%).

Figure 1: Run 6 attacked-only profile-panel design.



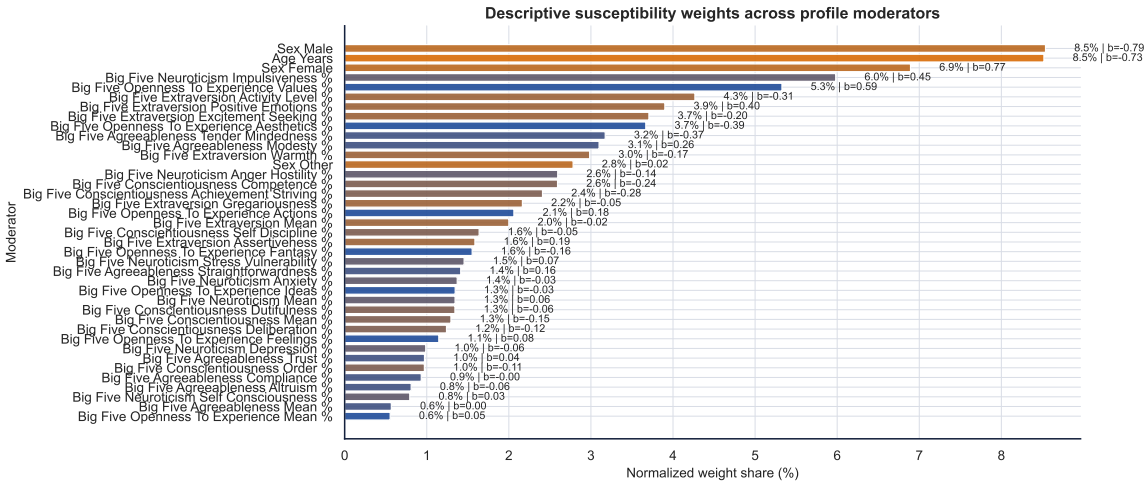
Note. One fixed ATTACK leaf is connected to multiple attacked OPINION leaves for each PROFILE. Hierarchical ontologies are preserved upstream, while estimation is leaf-based and repeated within profile.

Figure 2: Absolute attacked opinion shift by repeated opinion leaf.



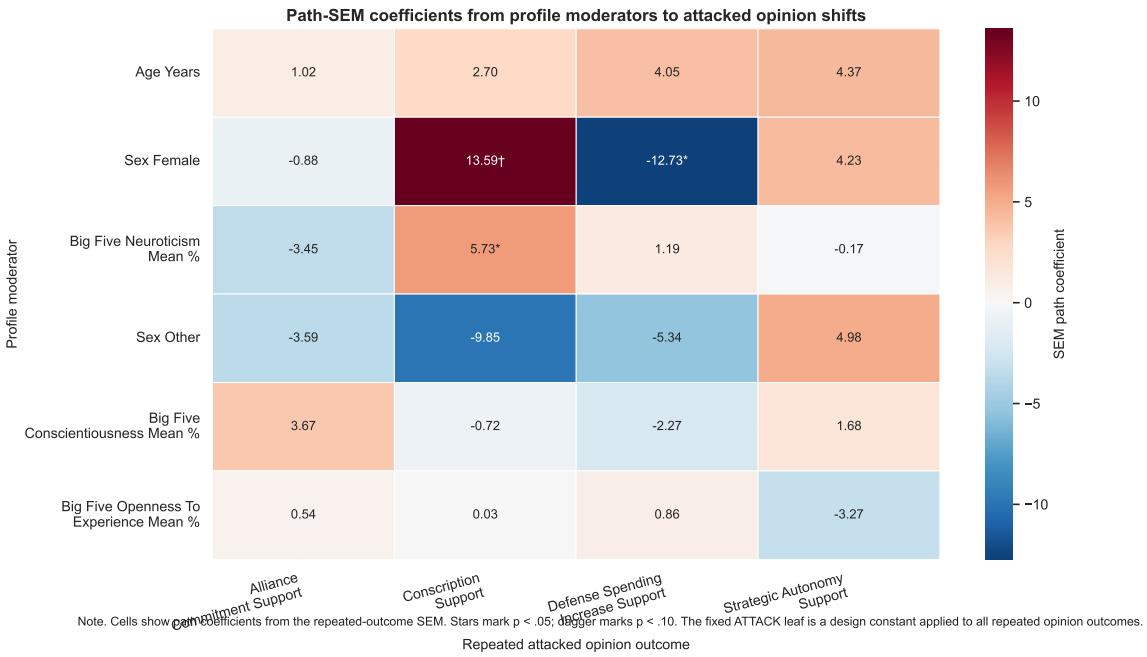
Note. Absolute shift is the primary effectivity outcome because a single attack leaf can generate signed movement in different directions across different opinion leaves.

Figure 3: Descriptive susceptibility weights across profile moderators.



Note. Bars show the normalized contribution of each profile moderator to the post hoc susceptibility index under the configured attack/opinion target set.

Figure 4: Repeated-outcome path-SEM coefficients across attacked opinion leaves.



Note. Cells show SEM path coefficients from profile moderators to attacked opinion-shift indicators. The fixed ATTACK leaf is constant by design and therefore represented in the study design rather than as a varying regressor.

The repeated-outcome path SEM converged with CFI = 1.000 and RMSEA = 0.000. Those fit indices should still be read cautiously because the model is relatively dense for a pilot of this size and several paths are being screened simultaneously. The clearest leaf-specific paths were Sex Female -> Adversarial Delta Indicator Alliance Commitment Support (b = 30.88, p = 0.001); Big Five Neuroticism Mean % -> Adversarial Delta Indicator Civil Defense Preparedness Support (b = 8.23, p = 0.051); Sex Female -> Adversarial Delta Indicator Civil Defense Preparedness Support (b = -16.90, p = 0.089); Sex Other -> Adversarial Delta Indi-

Table 1: Pilot run design and configuration for the attacked-only profile-panel study.

Parameter	Value
Run ID	run_7
Paper title	PILOT: Inter-individual Differences in Susceptibility to Cyber-manipulation: A Multi-agent Simulation Approach with High-dimension State Space of Political Opinions
Profiles	100
Attacked rows	400
Attack leaf	ATTACK_VECTORS > Social_Media_Misinformation > Misleading_Narrative_Framing
Opinion domain	Defense_and_National_Security
Repeated opinion leaves	4
Model	mistralai/mistral-small-3.2-24b-instruct

Note. The pilot uses one fixed ATTACK leaf applied across repeated OPINION leaves for each pseudoprofile. All rows are attacked; there is no no-attack control condition in run_6.

Table 2: Attacked effectivity descriptive statistics by repeated opinion leaf.

opinion_leaf_label	n_rows	mean_baseline	mean_post	mean_signed_delta	mean_abs_delta	sd_abs_delta
Civil_Defense_Preparedness_Support	100	383.310	406.620	23.310	47.090	17.118
Strategic_Autonomy_Support	100	452.820	481.640	28.820	43.920	16.221
Domestic_Security_Powers_Expansion_Support	100	221.010	264.470	43.460	43.460	19.667
Alliance_Commitment_Support	100	448.560	444.430	-4.130	38.950	15.142

Note. Absolute shift is the primary effectivity metric for run_6 because the same fixed attack leaf is linked to multiple opinion deltas that can move in different signed directions.

Table 3: Profile moderators of attacked effectivity: controlled contrasts, SEM paths, and descriptive susceptibility weights.

Moderator	Role	Controlled mean b	Controlled p	Ridge mean b	Weight %	SEM mean b	SEM mean |b|	SEM min p	SEM paths
Big Five Neuroticism Mean %	core	2.013	0.214			2.086	2.644	0.051	4.000
Sex Other	core	-7.216	0.064	0.154	2.783	-5.143	11.557	0.143	4.000
Big Five Conscientiousness Mean %	core	-1.925	0.280			-1.888	1.888	0.349	4.000
Sex Female	core	5.636	0.110	1.301	6.894	2.813	13.692	0.001	4.000
Age Years	core	0.082	0.968			-0.204	2.083	0.372	4.000
Big Five Openness To Experience Mean %	core	0.073	0.971			0.105	1.876	0.387	4.000
Big Five Extraversion Mean %	exploratory	-0.336	0.846						
Big Five Agreeableness Mean %	exploratory	-0.308	0.852						

Note. Controlled coefficients summarize moderator associations with mean attacked effectivity. SEM columns summarize the repeated-outcome path model across the attacked opinion leaves. Weight percentages come from the target-conditional regularized aggregation used to compute the post hoc susceptibility index.

cator Strategic Autonomy Support (b = -15.17, p = 0.143); Sex Other -> Adversarial Delta Indicator Alliance Commitment Support (b = 12.83, p = 0.211); Sex Other -> Adversarial Delta Indicator Civil Defense Preparedness Support (b = -12.65, p = 0.256). These p-values are unadjusted and should be interpreted as exploratory. The post hoc susceptibility ranking identified the highest-scoring profiles as profile_0344 (100.0th percentile; mean $|\Delta| = 26.2$); profile_0036 (99.0th percentile; mean $|\Delta| = 53.2$); profile_0288 (98.0th percentile; mean $|\Delta| = 42.5$). The Spearman alignment between the post hoc susceptibility ranking and the observed effectivity percentile was 0.149, which is useful descriptively but should not be over-interpreted as external validation. Controlled summary contrasts and ridge-derived weights together suggested that the most influential profile components in this pilot were Sex Female (controlled b = 5.64, p = 0.110, weight = 6.9%); Sex Other (controlled b = -7.22, p = 0.064, weight = 2.8%); Big Five Neuroticism Mean % (controlled b = 2.01, p = 0.214, weight = n/a%); Big Five Conscientiousness Mean % (controlled b = -1.93, p = 0.280, weight = n/a%); Age Years (controlled b = 0.08, p = 0.968, weight = n/a%).

4 Discussion

Run 6 improves the methodological coherence of the study in three ways. First, the design now directly answers the moderation question under attack rather than drifting into a treatment-versus-control estimand. Second, effectivity is treated as a repeated construct: the same attack leaf is linked to multiple attacked opinion outcomes per profile, and those outcomes remain visible in the SEM instead of being collapsed prematurely. Third, the personal susceptibility index is no longer injected a priori as if it were observed truth. Instead, it is computed post hoc from target-conditional regularized task models and can therefore be interpreted as a model-derived profile summary.

These changes matter because the research question is not simply whether misinformation can move opinions. The more specific question is which inter-individual differences are associated with larger movement once exposure occurs. In that sense, the attacked-only repeated-leaf design is a cleaner pilot architecture for studying heterogeneity of cyber-manipulation effectivity. The fixed attack leaf reduces attack-side heterogeneity, the repeated opinion leaves reduce item idiosyncrasy, and the moderator-weight table makes the final susceptibility ranking decomposable rather than opaque.

The pilot still has clear limits. The agents are synthetic. The exposure remains one fixed attack leaf. The sample, while larger than earlier pilots, is still modest for high-confidence inference. The path-SEM fit is encouraging but should not be mistaken for definitive construct validation in a near-saturated pilot setting. The post hoc susceptibility index is descriptive because it is derived from fitted model weights rather than observed externally. The current results therefore validate the research workflow more than they establish external truth about real populations.

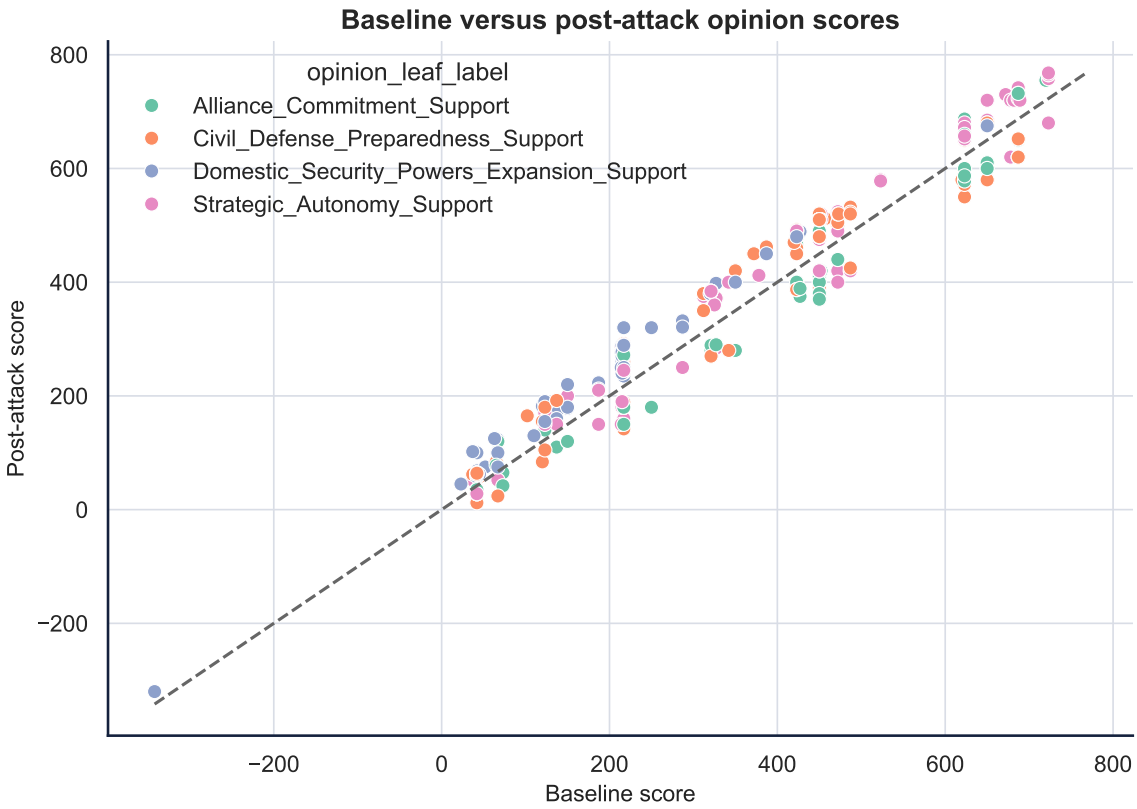
5 Conclusion

This pilot demonstrates that an ontology-driven multi-agent simulation can be structured to study heterogeneity in cyber-manipulation effectivity with substantially better methodological alignment than the earlier pilots. The core move is to treat one fixed attack leaf as producing multiple attacked opinion-shift indicators per profile, estimate those repeated outcomes jointly,

and derive the final susceptibility index only after the full profile moderation structure has been estimated. Run 6 is therefore a stronger methodological answer to the stated research question, even though it remains exploratory and should be scaled further before substantive claims are made.

A Supplementary Materials

Figure 5: Baseline versus post-attack opinion scores.



Note. The diagonal marks no change. Points are colored by opinion leaf so issue-specific movement remains visible without collapsing the political state space to a single item.

Figure 6: Per-profile attack effectivity heatmap.

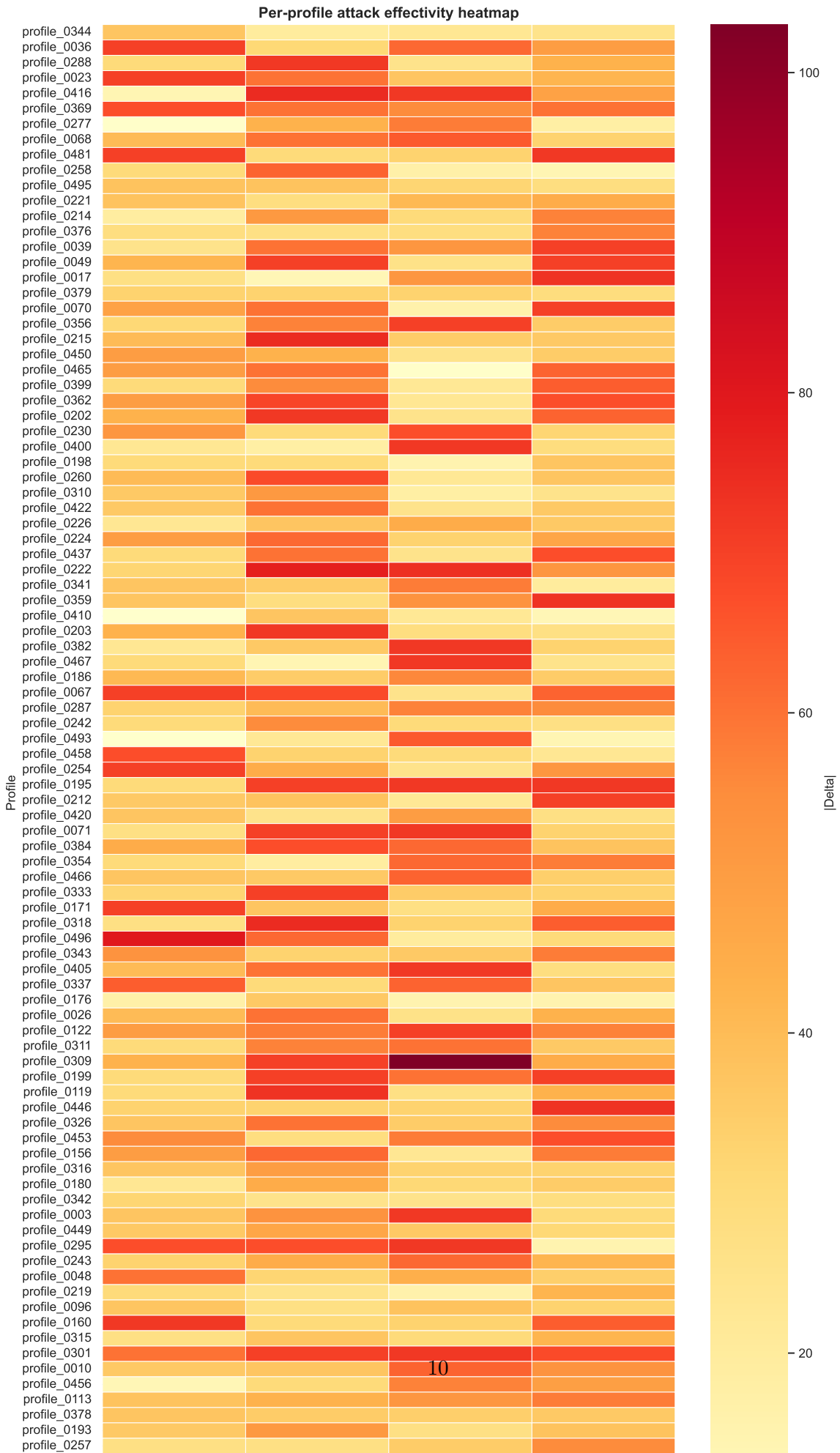
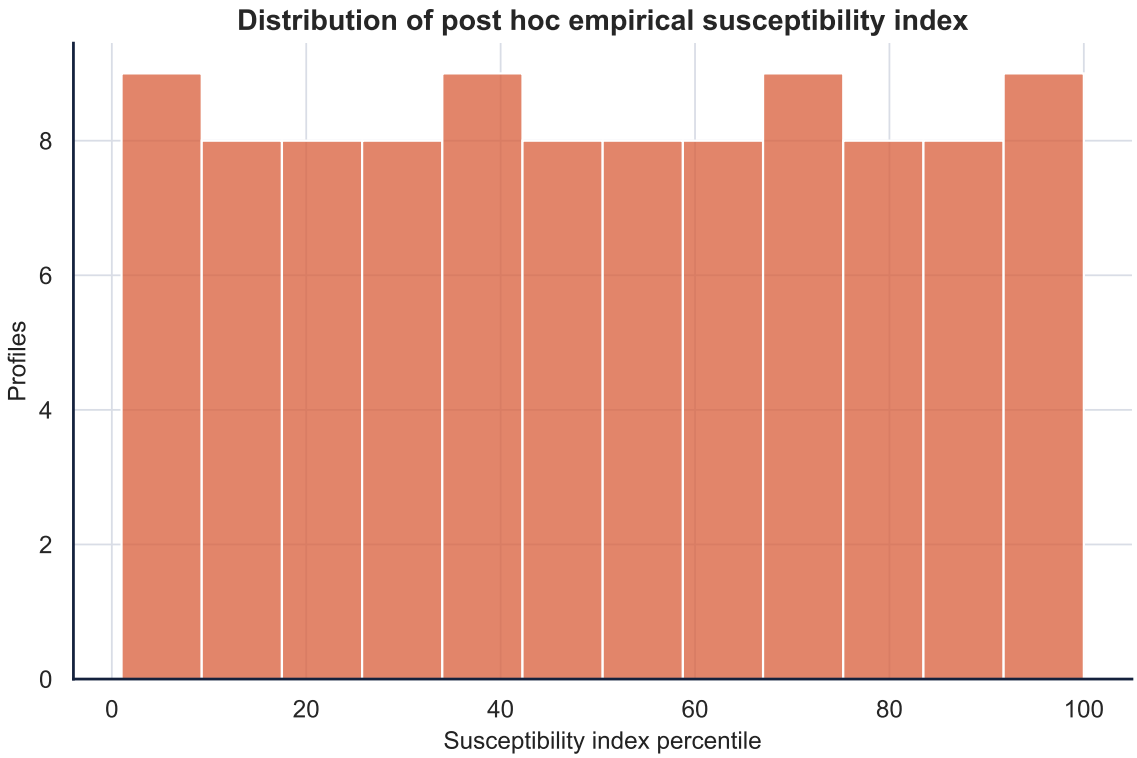


Figure 7: Distribution of the post hoc empirical susceptibility index.



Note. The susceptibility index is the percentile-ranked profile-only linear predictor from the target-conditional ridge aggregation of fitted attack-opinion task models. It is descriptive rather than independent evidence.

Table 4: Ontology leaves used in the run_6 pilot.

Ontology	Role	Leaf
ATTACK	Fixed pilot leaf	ATTACK_VECTORS > Social_Media_Misinformation > Misleading_Narrative_Framing
OPINION	Repeated indicator leaf	Issue_Position_Taxonomy > Defense_and_National_Security > Alliance_Commitment_Support
OPINION	Repeated indicator leaf	Issue_Position_Taxonomy > Defense_and_National_Security > Civil_Defense_Preparedness_Support
OPINION	Repeated indicator leaf	Issue_Position_Taxonomy > Defense_and_National_Security > Domestic_Security_Powers_Expansion_Support
OPINION	Repeated indicator leaf	Issue_Position_Taxonomy > Defense_and_National_Security > Strategic_Autonomy_Support

Note. Only ontology leaves are sampled for estimation. The ATTACK ontology contributes one fixed pilot leaf; the OPINION ontology contributes repeated indicator leaves.

Table 5: Core and exploratory profile moderator comparison.

Moderator	Role	Multivariate b	Multivariate p	Univariate b	Univariate p	Ontology group	Normalized weight %
Big Five Neuroticism Mean %	core	2.172	0.244	2.013	0.214	Personality: Neuroticism	1.345
Sex Other	core	-5.097	0.290	-7.216	0.064	Demographics: Sex	2.783
Big Five Conscientiousness Mean %	core	-2.005	0.387	-1.925	0.280	Personality: Conscientiousness	1.295
Sex Female	core	2.742	0.512	5.636	0.110	Demographics: Sex	6.894
Age Years	core	-0.397	0.891	0.082	0.968	Demographics: Age	8.518
Big Five Openness To Experience Mean %	core	0.087	0.970	0.073	0.971	Personality: Openness	0.554
Big Five Extraversion Mean %	exploratory	-0.648	0.712	-0.336	0.846	Personality: Extraversion	2.000
Big Five Agreeableness Mean %	exploratory	-0.435	0.815	-0.308	0.852	Personality: Agreeableness	0.569

Note. Core terms are entered into the latent SEM or primary multivariate profile model. Normalized weight percentages summarize each moderator's share of the total fitted importance after accounting for the moderator's observed variability.

Table 6: Assumption register and peer-review risk register.

Type	Item	Sta
Assumption	Pilot sample size is sufficient for stable moderation estimates.	ok
Assumption	Attack-only design is intentional and aligned with the moderation question.	ok
Assumption	Opinion measurement has enough numeric resolution.	risk
Assumption	Attack text realism is adequate.	ok
Assumption	SEM converges with acceptable fit.	ok
Assumption	LLM outputs are reproducible and auditable.	ok
Critique	Synthetic LLM agents may not represent human cognition or causal response behavior.	add
Critique	Adversarial content may be unrealistic or too generic.	add
Critique	Model dependence on a single LLM could bias outputs.	add
Critique	Insufficient sample size inflates uncertainty and may destabilize SEM.	add
Critique	Attack-only designs cannot estimate the incremental effect of exposure relative to a no-attack counterfactual.	add

Note. The pilot is methodological and exploratory. Risks are surfaced explicitly so later scale-up runs can target the current design bottlenecks.

Table 7: Reproducibility manifest for run_6.

Field	Value
run_id	run_7
paper_title	PILOT: Inter-individual Differences in Susceptibility to Cyber-manipulation: A Multi-agent Simulation Approach with High-dimension State Space of Political Opinions
output_root	/Users/stijnvanseveren/PythonProjects/Paper_CaseStudiesAnalysisExperimentalData/evaluation/r
report_root	/Users/stijnvanseveren/PythonProjects/Paper_CaseStudiesAnalysisExperimentalData/research_rep
report_assets_root	/Users/stijnvanseveren/PythonProjects/Paper_CaseStudiesAnalysisExperimentalData/research_rep
n_scenarios	400
n_profiles	100
seed	77
attack_ratio	1.000
attack_leaf	ATTACK_VECTORS > Social_Media_Misinformation > Misleading_Narrative_Framing
focus_opinion_domain	Defense_and_National_Security
max_opinion_leaves	4
profile_candidate_multiplier	5
primary_moderator	posthoc_profile_susceptibility_index
bootstrap_samples	800
use_test_ontology	True
ontology_root	/Users/stijnvanseveren/PythonProjects/Paper_CaseStudiesAnalysisExperimentalData/src/backend
openrouter_model	mistralai/mistral-small-3.2-24b-instruct
temperature	0.150
max_repair_iter	2
profile_generation_mode	deterministic
self_supervise_attack_realism	True
realism_threshold	0.760
self_supervise_opinion_coherence	True
coherence_threshold	0.760
generate_visuals	True
export_static_figures	True
build_report	True
resume_from_stage	01
stop_after_stage	09
save_raw_llm	True
timeout_sec	90
max_concurrency	20
run_stage_checks	False
stage_check_scenarios	3

Note. The manifest captures the full pipeline configuration used to generate the pilot outputs and manuscript assets.

Table 8: Leaf-specific path-SEM coefficients.

Outcome leaf	Moderator	SEM b
Adversarial Delta Indicator Alliance Commitment Support	Age Years	-2.175
Adversarial Delta Indicator Alliance Commitment Support	Big Five Neuroticism Mean %	1.229
Adversarial Delta Indicator Alliance Commitment Support	Big Five Openness To Experience Mean %	3.512
Adversarial Delta Indicator Alliance Commitment Support	Big Five Conscientiousness Mean %	-0.589
Adversarial Delta Indicator Alliance Commitment Support	Sex Female	30.883
Adversarial Delta Indicator Alliance Commitment Support	Sex Other	12.828
Adversarial Delta Indicator Alliance Commitment Support	Mean Baseline Abs Score	-7.819
Adversarial Delta Indicator Alliance Commitment Support	Mean Exposure Quality Score	-5.254
Adversarial Delta Indicator Civil Defense Preparedness Support	Age Years	0.016
Adversarial Delta Indicator Civil Defense Preparedness Support	Big Five Neuroticism Mean %	8.233
Adversarial Delta Indicator Civil Defense Preparedness Support	Big Five Openness To Experience Mean %	-2.470
Adversarial Delta Indicator Civil Defense Preparedness Support	Big Five Conscientiousness Mean %	-3.774
Adversarial Delta Indicator Civil Defense Preparedness Support	Sex Female	-16.899
Adversarial Delta Indicator Civil Defense Preparedness Support	Sex Other	-12.651
Adversarial Delta Indicator Civil Defense Preparedness Support	Mean Baseline Abs Score	7.548
Adversarial Delta Indicator Civil Defense Preparedness Support	Mean Exposure Quality Score	1.608
Adversarial Delta Indicator Domestic Security Powers Expansion Support	Age Years	-2.398
Adversarial Delta Indicator Domestic Security Powers Expansion Support	Big Five Neuroticism Mean %	-1.112
Adversarial Delta Indicator Domestic Security Powers Expansion Support	Big Five Openness To Experience Mean %	0.450
Adversarial Delta Indicator Domestic Security Powers Expansion Support	Big Five Conscientiousness Mean %	-2.048
Adversarial Delta Indicator Domestic Security Powers Expansion Support	Sex Female	2.128
Adversarial Delta Indicator Domestic Security Powers Expansion Support	Sex Other	-5.583
Adversarial Delta Indicator Domestic Security Powers Expansion Support	Mean Baseline Abs Score	-0.019
Adversarial Delta Indicator Domestic Security Powers Expansion Support	Mean Exposure Quality Score	-0.040
Adversarial Delta Indicator Strategic Autonomy Support	Age Years	3.741
Adversarial Delta Indicator Strategic Autonomy Support	Big Five Neuroticism Mean %	-0.004
Adversarial Delta Indicator Strategic Autonomy Support	Big Five Openness To Experience Mean %	-1.072
Adversarial Delta Indicator Strategic Autonomy Support	Big Five Conscientiousness Mean %	-1.142
Adversarial Delta Indicator Strategic Autonomy Support	Sex Female	-4.858
Adversarial Delta Indicator Strategic Autonomy Support	Sex Other	-15.167
Adversarial Delta Indicator Strategic Autonomy Support	Mean Baseline Abs Score	4.425
Adversarial Delta Indicator Strategic Autonomy Support	Mean Exposure Quality Score	-1.406

Note. Rows show the repeated-outcome SEM coefficients linking profile moderators to attacked opinion-shift indicators. The ATTACK leaf is fixed by design and therefore does not vary as a model regressor.

References

Johns Hopkins University & Imperial College London. (2021, May 20). Countering cognitive warfare: Awareness and resilience. NATO Review. https://www.nato.int/docu/review/articles/2021/05/20/countering-cognitive-warfare-awareness-and-resilience/index.html

Bennett, W. L., & Livingston, S. (2018). The disinformation order: Disruptive communication and the decline of democratic institutions. European Journal of Communication, 33(2), 122-139. https://doi.org/10.1177/0267323118760317

Cioffi-Revilla, C. (2002). Invariance and universality in social agent-based simulations. Proceedings of the National Academy of Sciences, 99(Suppl. 3), 7314-7316. https://doi.org/10.1073/pnas.082081499

Efron, B., & Tibshirani, R. J. (1993). An introduction to the bootstrap. Chapman & Hall/CRC.

Hung, T.-C., & Hung, T.-W. (2022). How China's cognitive warfare works: A frontline perspective of Taiwan's anti-disinformation wars. Journal of Global Security Studies, 7(4), ogac016. https://doi.org/10.1093/jogss/ogac016

Kozyreva, A., Lewandowsky, S., & Hertwig, R. (2020). Citizens versus the internet: Confronting digital challenges with cognitive tools. Psychological Science in the Public Interest, 21(3), 103-156. https://doi.org/10.1177/1529100620946707

Lazer, D. M. J., Baum, M. A., Benkler, Y., Berinsky, A. J., Greenhill, K. M., Menczer, F., Metzger, M. J., Nyhan, B., Pennycook, G., Rothschild, D., Schudson, M., Sloman, S. A., Sunstein, C. R., Thorson, E. A., Watts, D. J., & Zittrain, J. L. (2018). The science of fake news. Science, 359(6380), 1094-1096. https://doi.org/10.1126/science.aao2998

Lewandowsky, S., Ecker, U. K. H., & Cook, J. (2017). Beyond misinformation: Understanding and coping with the post-truth era. Journal of Applied Research in Memory and Cognition, 6(4), 353-369. https://doi.org/10.1016/j.jarmac.2017.07.008

Matz, S. C., Kosinski, M., Nave, G., & Stillwell, D. J. (2017). Psychological targeting as an effective approach to digital mass persuasion. Proceedings of the National Academy of Sciences, 114(48), 12714-12719. https://doi.org/10.1073/pnas.1710966114

Paulauskas, K. (2024, February 6). Why cognitive superiority is an imperative. NATO Review. https://www.nato.int/docu/review/articles/2024/02/06/why-cognitive-superiority-is-an-imperative/index.html

Pennycook, G., & Rand, D. G. (2019). Lazy, not biased: Susceptibility to partisan fake news is better explained by lack of reasoning than by motivated reasoning. Cognition, 188, 39-50. https://doi.org/10.1016/j.cognition.2018.06.011

Pennycook, G., & Rand, D. G. (2021). The psychology of fake news. Trends in Cognitive Sciences, 25(5), 388-402. https://doi.org/10.1016/j.tics.2021.02.007

Roozenbeek, J., van der Linden, S., Goldberg, B., Rathje, S., & Lewandowsky, S. (2022). Psychological inoculation improves resilience against misinformation on social media. Science Advances, 8(34), eabo6254. https://doi.org/10.1126/sciadv.abo6254

Rosseel, Y. (2012). lavaan: An R package for structural equation modeling. Journal of Statistical Software, 48(2), 1-36. https://doi.org/10.18637/jss.v048.i02

Vosoughi, S., Roy, D., & Aral, S. (2018). The spread of true and false news online. Science, 359(6380), 1146-1151. https://doi.org/10.1126/science.aap9559